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AN ATOM GENERALLY HAS MANY DIFFERENT ENERGY LEVELS. IN MANY EXPERIMENTS, THOUGH, ONLY **TWO** ENERGY LEVELS - USUALLY THE GROUND **STATE** AND ONE EXCITED STATE - PLAY A SIGNIFICANT ROLE

E_1 ————— $|E_1\rangle$ EXCITED STATE

E_0 ————— $|E_0\rangle$ GROUND STATE

$$\langle E_0|E_0\rangle = \langle E_1|E_1\rangle = 1 \quad \langle E_0|E_1\rangle = \langle E_1|E_0\rangle = 0$$

THE PRINCIPLE OF SUPERPOSITION TELLS US THAT THE OTHER STATES CAN BE FORMED BY COMPLEX LINEAR COMBINATIONS LIKE

$$|\psi\rangle = \alpha_0|E_0\rangle + \alpha_1|E_1\rangle$$

$$|\alpha_0|^2 = \text{PROB. OF FINDING THE ATOM IN THE STATE } |E_0\rangle$$

$$|\alpha_1|^2 = \text{PROB. OF FINDING THE ATOM IN THE STATE } |E_1\rangle$$

$$|\alpha_0|^2 + |\alpha_1|^2 = 1$$

SHOW THAT ANY STATE OF THE FORM $e^{i\phi}|E_k\rangle$ IS A STATE WITH DEFINITE ENERGY E_k

• IN STATE $|E_k\rangle$ THE ENERGY IS E_k

$$|\psi\rangle = e^{i\phi}|E_k\rangle$$

$$P_{E_k} = |\langle E_k|\psi\rangle|^2 = |\langle E_k|e^{i\phi}|E_k\rangle|^2 = |e^{i\phi}\underbrace{\langle E_k|E_k\rangle}_1|^2 = \underbrace{|e^{i\phi}|^2}_{\text{EULER'S IDENTITY}} = 1$$

• A MEASUREMENT OF ENERGY ON THE TWO-LEVEL ATOM IS ASSOCIATED WITH THE TWO ENERGY LEVEL STATES $|E_0\rangle$ AND $|E_1\rangle$

• MEASUREMENTS OF OTHER VARIABLES WILL BE ASSOCIATED WITH OTHER ORTHONORMAL PAIRS OF STATES. SUPPOSE $|u\rangle$ IS THE STATE ASSOCIATED WITH SOME MEASUREMENT OUTCOME u

• THIS IS SIMILAR TO WHAT WE HAVE SEEN FOR THE SPIN-1/2 PARTICLE! THINK ABOUT

$$|y_+\rangle = \frac{1}{\sqrt{2}}|z_+\rangle + \frac{1}{\sqrt{2}}|z_-\rangle \text{ AND MEASURING } S_y$$

MEANS THAT
MEASUREMENT
 $\downarrow \downarrow$
 $|\langle x_+|x_+\rangle|^2 = 1$

$$S_y \begin{cases} +\hbar/2 \rightarrow |y_+\rangle = \frac{1}{\sqrt{2}}(|z_+\rangle + i|z_-\rangle) \\ -\hbar/2 \rightarrow |y_-\rangle \end{cases}$$

$$\langle y_+| = \frac{1}{\sqrt{2}} \langle z_+| - i \langle z_-|$$

TRANSPOSE AND COMPLEX CONJUGATE

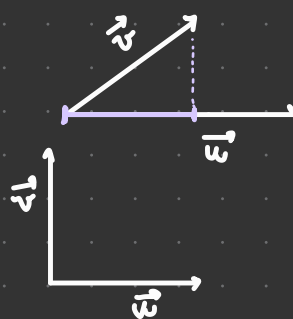
$$\begin{aligned} P_y(+\hbar/2) &= |\langle y_+|x_+\rangle|^2 = \left| \frac{1}{\sqrt{2}} (\langle z_+| - i \langle z_-|) \frac{1}{\sqrt{2}} (|z_+\rangle + |z_-\rangle) \right|^2 \\ &= \frac{1}{4} \left| \underbrace{\langle z_+|z_+\rangle}_1 + \underbrace{\langle z_+|z_-\rangle}_0 - i \underbrace{\langle z_-|z_+\rangle}_1 - i \underbrace{\langle z_-|z_-\rangle}_1 \right|^2 \\ &= \frac{1}{4} |(1-i)|^2 = \frac{1}{4} (1+i) = \frac{1}{2} \end{aligned}$$

SUPPOSE THAT THE ATOM IS IN THE STATE $|\psi\rangle = \alpha_0|E_0\rangle + \alpha_1|E_1\rangle$

AND WE WANT TO CALCULATE THE PROBABILITY THAT THE MEASUREMENT OF A CERTAIN PROPERTY GIVES THE OUTCOME w

WE KNOW THAT

SCALAR PRODUCT



$$\vec{v} \cdot \vec{w} = x_v x_w + y_v y_w$$

$$\begin{pmatrix} x_w \\ y_w \end{pmatrix}$$

$$\langle v | w \rangle = \alpha_v^* \alpha_w + \beta_v^* \beta_w$$

HOW DOES A QUANTUM SYSTEM EVOLVE WITH TIME?

WE CONSIDER AN INFORMATIONALLY ISOLATED ATOM IN A CERTAIN INITIAL STATE $|\psi(0)\rangle = |E_n\rangle$

HAVING ENERGY E_n AT TIME $t=0$

IT TURNS OUT THAT THE STATE EVOLVES IN TIME ACCORDING TO

$$|\psi(t)\rangle = e^{i\frac{E_n}{\hbar}t} |E_n\rangle = e^{-i\omega_n t} |E_n\rangle$$

DOES IT MAKE ANY SENSE?

$$|\psi(t)\rangle = e^{-i\frac{E_n}{\hbar}t} |E_n\rangle = e^{-i\omega_n t} |E_n\rangle$$

$$|\psi(0)\rangle = |E_n\rangle$$

THE (INFORMATIONALLY ISOLATED) ATOM HAS ENERGY E_n AT ANY TIME

$$P_{E_n} = |\langle E_n | \psi(t) \rangle|^2 = 1$$

IN FACT THE PROB. OF ANY OUTCOME w STAYS THE SAME

$$P_w(0) = |\langle w | \psi(0) \rangle|^2 = |\langle w | E_n \rangle|^2$$

$$P_w(t) = |\langle w | \psi(t) \rangle|^2 = |\langle w | e^{-i\omega_n t} | E_n \rangle|^2 =$$

$$= |e^{-i\omega_n t} \langle w | E_n \rangle|^2 = \underbrace{|e^{i\omega_n t}|^2}_{=1} |\langle w | E_n \rangle|^2 = |\langle w | E_n \rangle|^2 = P_w(0)$$

THE STATES LIKE $|E_n\rangle$ ARE CALLED STATIONARY

WHAT HAPPENS IF THE STATE IS IN A SUPERPOSITION OF STATIONARY STATES?

$$|\psi\rangle = \alpha_0|E_0\rangle + \alpha_1|E_1\rangle$$

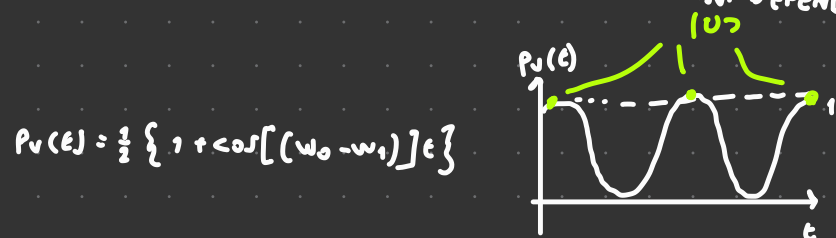
$$\begin{aligned}
 P_{E_1}(t) &= |\langle E_1 | \psi(t) \rangle|^2 = |\langle E_1 | a_0 e^{-i\omega_0 t} | E_0 \rangle + a_1 e^{-i\omega_1 t} | E_1 \rangle|^2 \\
 &= a_0 e^{-i\omega_0 t} \underbrace{\langle E_1 | E_0 \rangle}_0 + a_1 e^{-i\omega_1 t} \underbrace{\langle E_1 | E_1 \rangle}_1 \\
 &= |a_1 e^{-i\omega_1 t}|^2 = |a_1|^2
 \end{aligned}$$

• LET US CONSIDER THE STATE $|V\rangle$ ASSOCIATED WITH A CERTAIN OUTCOME v WITH $|V\rangle = \frac{1}{\sqrt{2}} |E_0\rangle + \frac{1}{\sqrt{2}} |E_1\rangle$

• WE ASSUME THAT OUR SYSTEM IS INITIALLY (I.E. $t=0$) IN THE STATE $|\psi(0)\rangle = |V\rangle$

$$\begin{aligned}
 P_V(0) &= ? \\
 P_V(t) &= ? \\
 |\psi(t)\rangle &= \frac{1}{\sqrt{2}} e^{-i\omega_0 t} |E_0\rangle + \frac{1}{\sqrt{2}} e^{-i\omega_1 t} |E_1\rangle \\
 |\langle V | \psi(t) \rangle|^2 &= \left| \frac{1}{\sqrt{2}} \langle E_0 | + \frac{1}{\sqrt{2}} \langle E_1 | \right| \left(\frac{1}{\sqrt{2}} e^{-i\omega_0 t} |E_0\rangle + \frac{1}{\sqrt{2}} e^{-i\omega_1 t} |E_1\rangle \right) \right|^2 \\
 &= \left| \frac{1}{2} e^{-i\omega_0 t} + \frac{1}{2} e^{-i\omega_1 t} + 0 + 0 \right|^2 = \frac{1}{4} (e^{-i\omega_0 t} + e^{-i\omega_1 t})(e^{i\omega_0 t} + e^{i\omega_1 t}) \\
 &= \frac{1}{4} (1 + e^{-i(\omega_1 - \omega_0)t} + e^{i(\omega_0 - \omega_1)t} + 1) \quad e^{ix} + e^{-ix} = 2\cos x \\
 &= \frac{1}{4} (2 + 2\cos[(\omega_1 - \omega_0)t]) = \frac{1}{2} (1 + \cos[(\omega_0 - \omega_1)t])
 \end{aligned}$$

→ RANGES BETWEEN 0 AND 1
BUT DEPENDS ON TIME



THE PROBABILITY $P_V(t)$ OSCILLATES WITH THE ANGULAR FREQUENCY $\omega_0 - \omega_1$ AT LEAST SOME OF THE OBSERV

$$P_V = |\langle V | \psi(t) \rangle|^2$$

$$|V\rangle |V_L\rangle \langle V_L | V \rangle = 0$$

WE HAVE POSTULATED THAT THE TIME EVOLUTION OF AN INFORMATIONALLY ISOLATED TWO LEVEL ATOM IS LINEAR

$$\alpha |\psi(0)\rangle + \beta |\varphi(0)\rangle \rightarrow \alpha |\psi(t)\rangle + \beta |\varphi(t)\rangle$$

WE CAN ALSO SAY THAT THE TIME EVOLUTION OF THE SYSTEM IS DESCRIBED BY A **LINEAR OPERATOR** ACTING ON THE INITIAL STATE

$|\psi\rangle \rightarrow A|\psi\rangle$ THE STATE $|\psi\rangle$ IS MAPPED INTO A NEW STATE BY THE OPERATOR A

THE TIME-EVOLUTION OPERATOR MUST SATISFY A FEW PROPERTIES

1) $U(t)|E_n\rangle = e^{-iE_n t/\hbar} |E_n\rangle$ FOR AN ENERGY STATE

2) $U(t)$ MUST BE LINEAR

3) SINCE EVERY STATE IS NORMALIZED $U(t)$ MUST BE UNITARY

$$|\psi(t)\rangle = U(t)|\psi(0)\rangle$$

NOTE THAT IF WE KNOW THE TIME EVOLUTION OPERATOR FOR ALL t , THEN WE KNOW

THE PRODUCT AB OF TWO OPERATORS A AND B IS THE OPERATOR THAT ACTS THIS WAY:

$$AB|\psi\rangle = A(B|\psi\rangle) \text{ FOR ANY } |\psi\rangle$$

$$|\psi\rangle = 2\alpha|\epsilon_0\rangle + \alpha_1|\epsilon_1\rangle$$

$$U|\epsilon_1\rangle$$

WE INTRODUCE THE ENERGY